

DSS Application Design Document

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1 Overview

This is a design document for a distributed storage system (DSS). It includes the message formats for the implemented commands, the time-space diagrams, the data structures and design decisions, the port numbers, the screenshots, the video demo link and timestamps.

2 Message Formats for Implemented Commands

This milestone uses a common message envelope for all requests and responses. Individual commands define their own bodies and semantics.

2.1 Envelope Format (Common to All Messages)

Envelope

```
{
  "version": 1,
  "message_id": "<role>-<pid>-<timestamp_ms>-<counter>",
  "message_type": "<command_or_response>",
  "status_code": "SUCCESS | FAILURE (optional on requests)",
  "reason": "Optional explanation on FAILURE",
  "body": { /* command-specific fields */ }
}
```

Message ID Format

Pattern <role>-<pid>-<timestamp_ms>-<counter> (e.g., user1-4123-1732665001123-00042).

Role: user name or disk name (e.g., *manager*, *user1*, *diskA*)

PID: process ID of the sender

Timestamp: current time in milliseconds

Counter: incremented each time this process sends a message

Purpose: match requests and responses, enable retries on UDP, deduplicate requests, and simplify logging.

2.2 Command: register_user

Purpose

Register a user process with the manager.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user",
  "body": {
    "user_name": "User1",
    "ipv4_address": "192.0.2.10",
    "management_port": 2101,
    "command_port": 2102
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user_response",
  "status_code": "SUCCESS",
  "reason": null
}
```

Time-Space Diagram

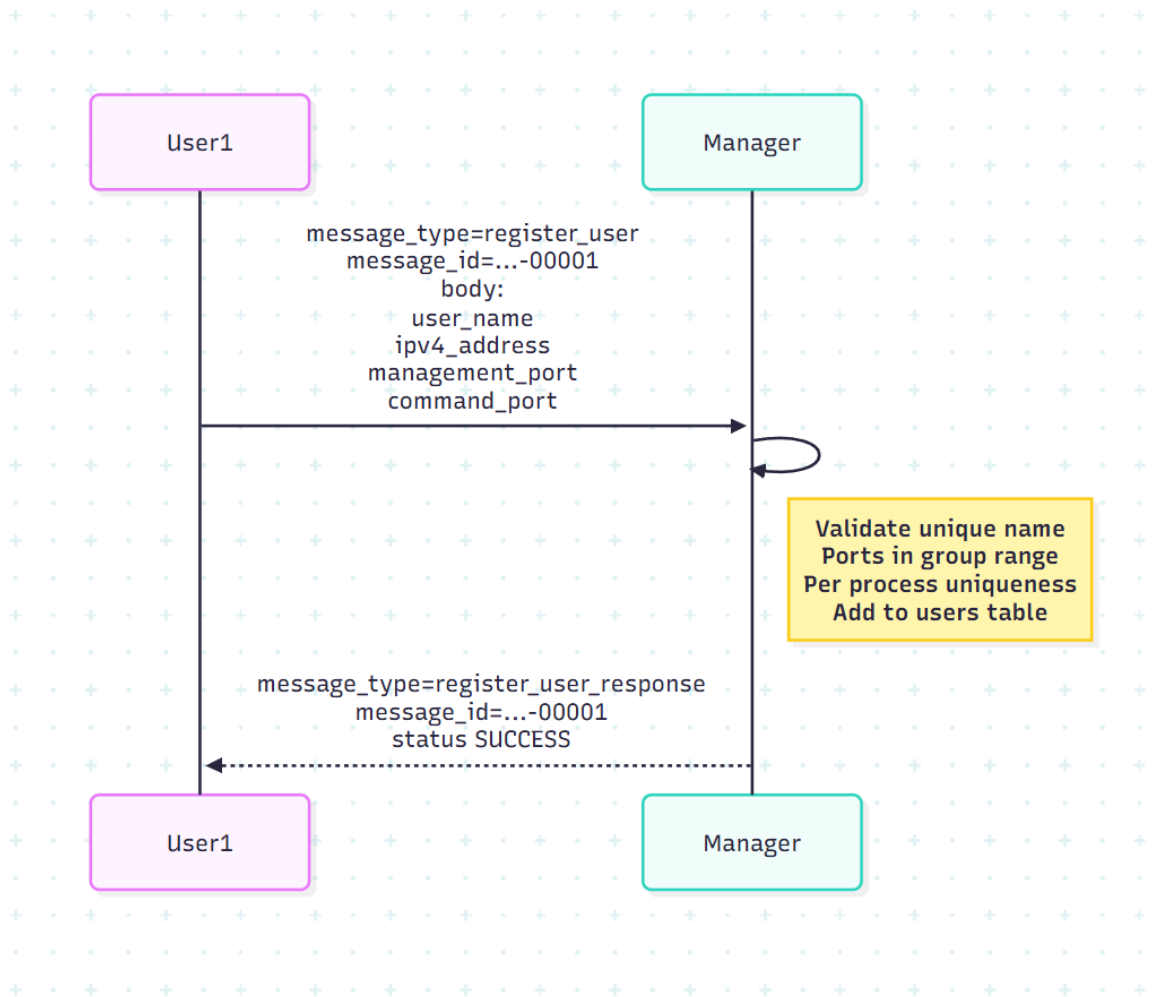


Figure 1: Time-Space diagram for `register_user`

2.3 Command: register_disk

Purpose

Register a disk process with the manager.

Request

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk",
  "body": {
    "disk_name": "DiskA",
    "ipv4_address": "192.0.2.11",
    "management_port": 2201,
    "command_port": 2202
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk_response",
  "status_code": "SUCCESS"
}
```

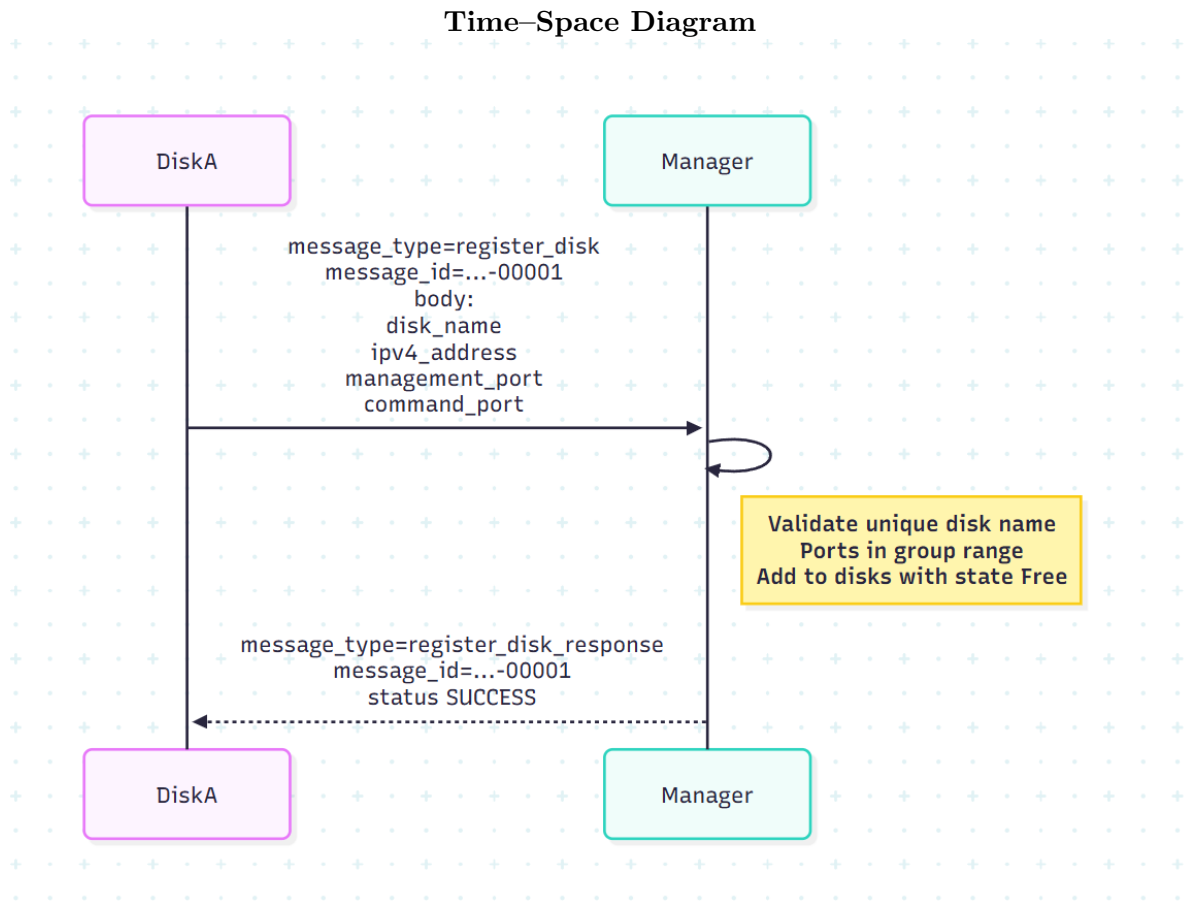


Figure 2: Time-space diagram for `register_disk`

2.4 Command: configure_dss

Purpose

Configure a DSS from available disks.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss",
  "body": {
    "dss_name": "DSS1",
    "number_of_disks": 3,
    "striping_unit_bytes": 1024
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss_response",
  "status_code": "SUCCESS",
  "body": {
    "dss_name": "DSS1",
    "disk_order": ["DiskA", "DiskB", "DiskC"],
    "striping_unit_bytes": 1024
  }
}
```

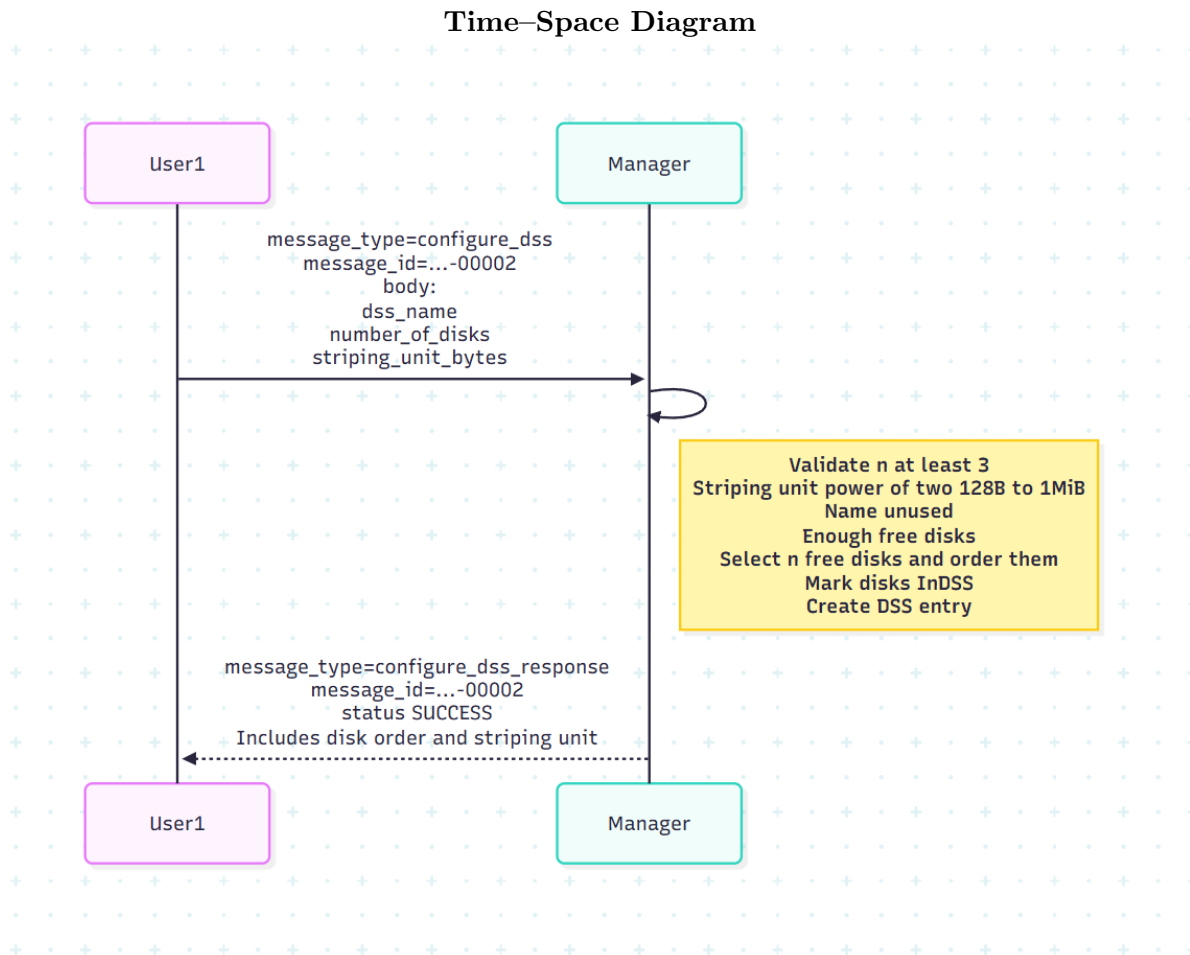


Figure 3: Time-space diagram for `configure_dss`

2.5 Command: deregister_user

Purpose

Deregister a user process from the manager.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002500-00003",
  "message_type": "deregister_user",
  "body": {
    "user_name": "User1"
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002500-00003",
  "message_type": "deregister_user_response",
  "status_code": "SUCCESS"
}
```

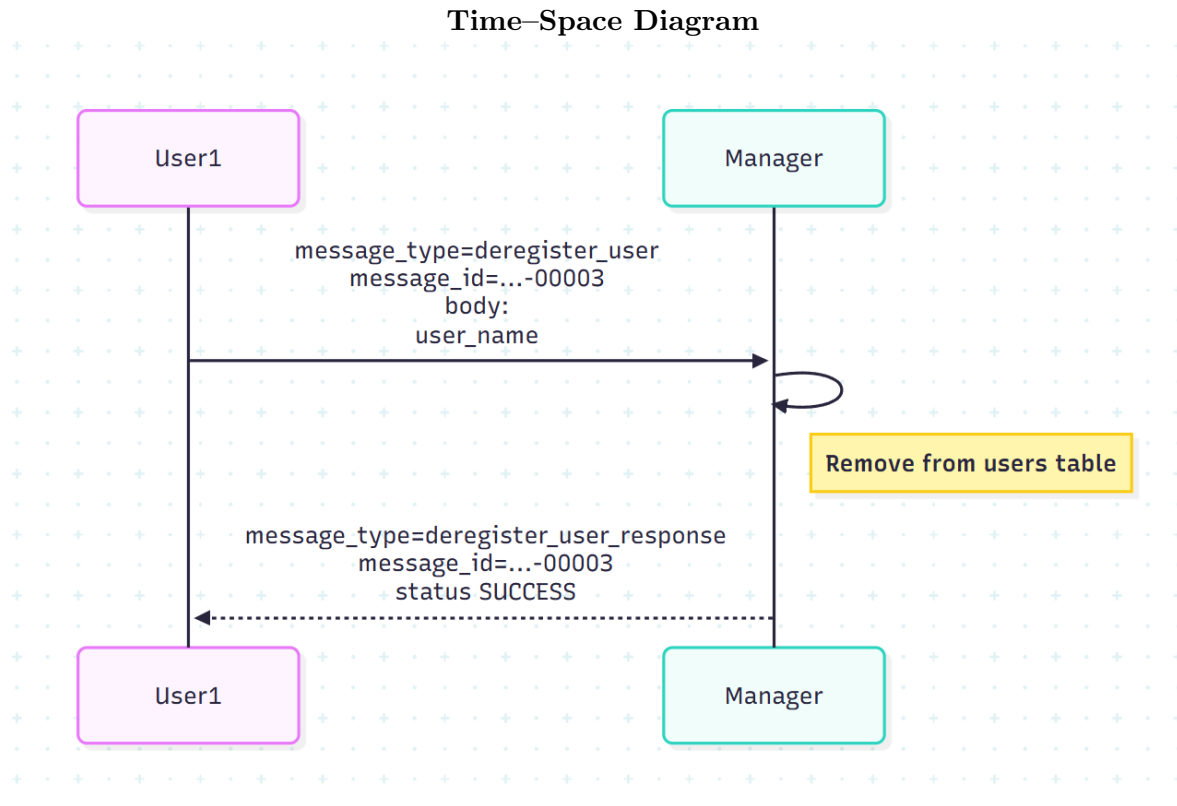


Figure 4: Time-space diagram for `deregister_user`

2.6 Command: deregister_disk

Purpose

Deregister a disk process from the manager.

Request

```
{  
  "version": 1,  
  "message_id": "diskA-6123-1732665002700-00004",  
  "message_type": "deregister_disk",  
  "body": {  
    "disk_name": "DiskA"  
  }  
}
```

Response

```
{  
  "version": 1,  
  "message_id": "diskA-6123-1732665002700-00004",  
  "message_type": "deregister_disk_response",  
  "status_code": "SUCCESS"  
}
```

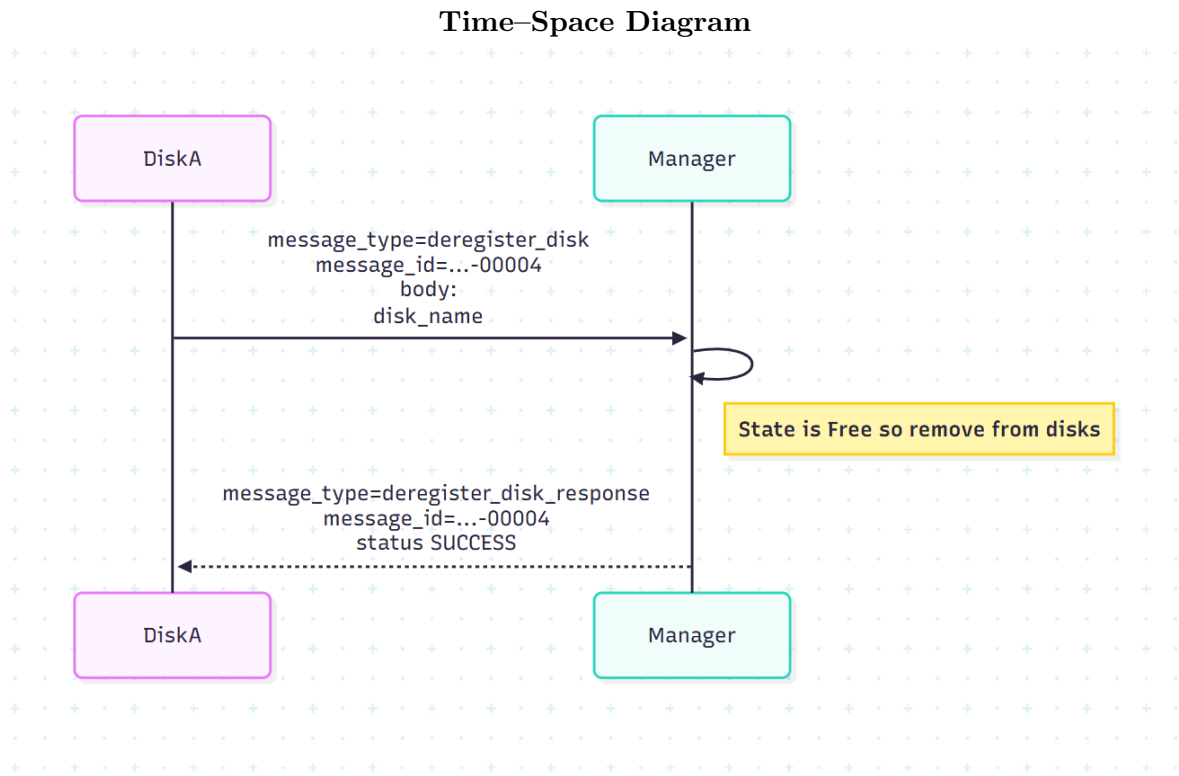


Figure 5: Time-space diagram for `deregister_disk`

3 Design Decisions

Key design decisions include:

- Using a common message envelope for all messages.
- Using a shared message ID pattern for all messages.
- Using standardized response/request message formats for all commands.

4 Port Numbers

Because I am Group 198, the port numbers are 21200–21299 (inclusive).

5 Screenshots

```
parker@DESKTOP-69P9G1L:/mnt/c/Users/pconley1/Dropbox/1. Projects/CSE 434/socket-project/Group198$ git log --pretty=format:"%h - %an, %ad (Commit) - %cd (Author)"
23359c6 - parkerconley, Fri Sep 26 18:22:08 2025 -0700 (Commit) - Fri Sep 26 18:22:08 2025 -0700 (Author)
9e7a297 - parkerconley, Fri Sep 26 16:31:25 2025 -0700 (Commit) - Fri Sep 26 16:31:25 2025 -0700 (Author)
d5d8694 - parkerconley, Fri Sep 26 16:28:23 2025 -0700 (Commit) - Fri Sep 26 16:28:23 2025 -0700 (Author)
0e5176f - parkerconley, Fri Sep 26 15:39:36 2025 -0700 (Commit) - Fri Sep 26 15:39:36 2025 -0700 (Author)
177745a - parkerconley, Fri Sep 26 14:46:58 2025 -0700 (Commit) - Fri Sep 26 14:46:58 2025 -0700 (Author)
3a3cab2 - parkerconley, Fri Sep 26 13:57:43 2025 -0700 (Commit) - Fri Sep 26 13:57:43 2025 -0700 (Author)
90547a4 - parkerconley, Fri Sep 26 12:51:36 2025 -0700 (Commit) - Fri Sep 26 12:51:36 2025 -0700 (Author)
a545e6e - parkerconley, Thu Sep 25 19:16:22 2025 -0700 (Commit) - Thu Sep 25 19:16:22 2025 -0700 (Author)
db66c92 - parkerconley, Thu Sep 25 18:07:28 2025 -0700 (Commit) - Thu Sep 25 18:07:28 2025 -0700 (Author)
a4296eb - parkerconley, Thu Sep 25 17:59:53 2025 -0700 (Commit) - Thu Sep 25 17:59:53 2025 -0700 (Author)
e901420 - parkerconley, Thu Sep 25 16:13:18 2025 -0700 (Commit) - Thu Sep 25 16:13:18 2025 -0700 (Author)
```

Figure 6: git log output

I didn't include the GitHub commit history due to it being paywalled by GitHub Plus.

Figure 7: Repository commit history

```
parker@DESKTOP-69P9G1L:/mnt/c/Users/pconley1/Dropbox/1. Projects/CSE 434/socket-project/Group198$ git reflog
23359c6 (HEAD -> main) HEAD@{0}: commit: finished design doc
9e7a297 HEAD@{1}: commit: readme update
d5d8694 HEAD@{2}: commit: mostly finished
0e5176f HEAD@{3}: commit: added register-disk
177745a HEAD@{4}: commit: added user.py register-user functionality, seems to work after a brief test
3a3cab2 HEAD@{5}: commit: initial implementation of register-user in manager.py
90547a4 HEAD@{6}: commit: created design doc
a545e6e HEAD@{7}: commit: added design doc
db66c92 HEAD@{8}: commit: test
a4296eb HEAD@{9}: commit: initial file structure
e901420 (origin/main, origin/HEAD) HEAD@{10}: clone: from github.com:parconley/Group198.git
```

Figure 8: git reflog output

```
parker@DESKTOP-69P9G1L:/mnt/c/Users/pconley1/Dropbox/1. Projects/CSE 434/socket-project/Group198$ git fsck
Checking object directories: 100% (256/256), done.
Checking objects: 100% (3/3), done.
```

Figure 9: git fsck output

The figures above correspond to:

- i. Output of: `git log --pretty=format:"%h - %an, %ad (Commit) - %cd (Author)"`
- ii. Entire commit history of the GitHub repository (omitted due to paywall)
- iii. Output of: `git reflog`
- iv. Output of: `git fsck`

6 Video Demo Link and Timestamps

Accessible Link <https://youtu.be/xnzJNYWxVww>

Timestamps

1. (a): N/A - Didn't need to compile.
2. (b): 0:00 - You can see that I have a CloudLab VM and personal PC running.
3. (c): 0:00 - I setup the manager, disks, and users.
4. (d): 2:30 - I configure the DSS.
5. (e): 3:28 - I try to configure the DSS again, but it fails due to insufficient disks.
6. (f): 4:03 - I deregister the disks and users, though the disks cannot deregister because they are in use.

7 Extended Design for Full Project

For the next version of this project, I'll need to add commands, add threading, and record another video demo.